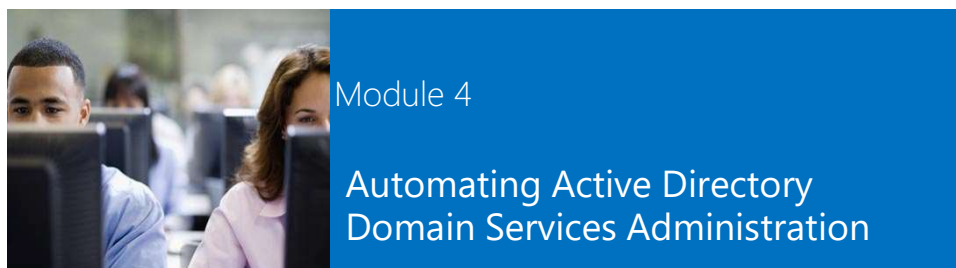


Microsoft® Official Course



Module Overview

- Using Command-line Tools for AD DS Administration
- Using Windows PowerShell for AD DS Administration
- Performing Bulk Operations with Windows PowerShell

Lesson 1: Using Command-line Tools for AD DS Administration

- Benefits of Using Command-Line Tools for AD DS Administration
- What Is Csvde?
- Working with CSV Files
- What Is Ldifde?
- What Are DS Commands?

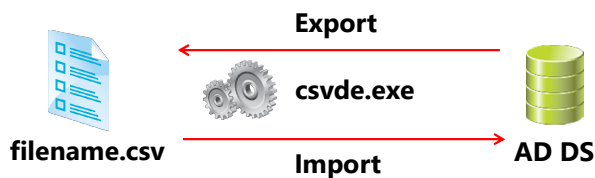
Benefits of Using Command-Line Tools for AD DS Administration

Command-line tools allow you to automate AD DS administration

Benefits of using command-line tools:

- Faster implementation of bulk operations
- Customized processes for AD DS administration
- AD DS administration on server core

What Is Csvde?



Use csvde to export objects to a .csv file:

- -f filename
- -d RootDN
- -p SearchScope
- -r Filter
- -l ListOfAttributes

Use csvde to create objects from a .csv file:

```
csvde -i -f filename -k
```

Working with CSV Files

The first line of a .csv file defines the names of the columns

```

FirstName,LastName,Department
Greg,Guzik,IT
Robin,Young,Research
Qiong,Wu,Marketing
  
```

A **foreach** loop processes the contents of a .csv that have been imported into a variable

```

$users=Import-CSV -LiteralPath "C:\users.csv"
foreach ($user in $users) {
    Write-Host "The first name is:"
    $user.FirstName
}
  
```

What Is Ldifde?



Use Ldifde to export objects to a LDIF file:

- -f filename
- -d RootDN
- -r Filter
- -p SearchScope
- -l ListOfAttributesToInclude
- -o ListOfAttributesToExclude

Use Ldifde to create, modify, or delete objects:

```
ldifde -i -f filename -k
```

What Are DS Commands?

Windows Server 2012 includes **ds*** commands that are suitable for use in scripts

- Examples
- To modify the department of a user account, type:

```
Dsmod user "cn=Joe Healy,ou=Managers,  
dc=adatum,dc=com" -dept IT
```

- To display the email of a user account, type:

```
Dsget user "cn=Joe Healy,ou=Managers,  
dc=adatum,dc=com" -email
```

- To delete a user account, type:

```
Dsrm "cn=Joe Healy,ou=Managers,dc=adatum,dc=com"
```

- To create a new user account, type:

```
Dsadd user "cn=Joe Healy,ou=Managers,dc=adatum,dc=com"
```

Lesson 2: Using Windows PowerShell for AD DS Administration

- Using Windows PowerShell Cmdlets to Manage User Accounts
- Using Windows PowerShell Cmdlets to Manage Groups
- Using Windows PowerShell Cmdlets to Manage Computer Accounts
- Using Windows PowerShell Cmdlets to Manage OUs
- Querying Objects with Windows PowerShell
- Modifying Objects with Windows PowerShell

Using Windows PowerShell Cmdlets to Manage User Accounts

Cmdlet	Description
New-ADUser	Creates user accounts
Set-ADUser	Modifies properties of user accounts
Remove-ADUser	Deletes user accounts
Set-ADAccountPassword	Resets the password of a user account
Set-ADAccountExpiration	Modifies the expiration date of a user account
Unlock-ADAccount	Unlocks a user account after it has become locked after too many incorrect login attempts
Enable-ADAccount	Enables a user account
Disable-ADAccount	Disables a user account

```
New-ADUser "Sten Faerch" -AccountPassword (Read-Host  
-AsSecureString "Enter password") -Department IT
```

Using Windows PowerShell Cmdlets to Manage Groups

Cmdlet	Description
New-ADGroup	Creates new groups
Set-ADGroup	Modifies properties of groups
Get-ADGroup	Displays properties of groups
Remove-ADGroup	Deletes groups
Add-ADGroupMember	Adds members to groups
Get-ADGroupMember	Displays membership of groups
Remove-ADGroupMember	Removes members from groups
Add-ADPrincipalGroupMembership	Adds group membership to objects
Get-ADPrincipalGroupMembership	Displays group membership of objects
Remove-ADPrincipalGroupMembership	Removes group membership from an object

```
New-ADGroup -Name "CustomerManagement" -Path  
"ou=managers,dc=adatum,dc=com" -GroupScope Global  
-GroupCategory Security
```

```
Add-ADGroupMember -Name "CustomerManagement"  
-Members "Joe"
```

Using Windows PowerShell Cmdlets to Manage Computer Accounts

Cmdlet	Description
New-ADComputer	Creates new computer accounts
Set-ADComputer	Modifies properties of computer accounts
Get-ADComputer	Displays properties of computer accounts
Remove-ADComputer	Deletes computer accounts
Test-ComputerSecureChannel	Verifies or repairs the trust relationship between a computer and the domain
Reset-ComputerMachinePassword	Resets the password for a computer account

```
New-ADComputer -Name "LON-SVR8" -Path  
"ou=marketing,dc=adatum,dc=com" -Enabled $true
```

```
Test-ComputerSecureChannel -Repair
```

Using Windows PowerShell Cmdlets to Manage OUs

Cmdlet	Description
New-ADOrganizationalUnit	Creates OUs
Set-ADOrganizationalUnit	Modifies properties of OUs
Get-ADOrganizationalUnit	Views properties of OUs
Remove-ADOrganizationalUnit	Deletes OUs

```
New-ADOrganizationalUnit -Name "Sales"
-Path "ou=marketing,dc=adatum,dc=com"
-ProtectedFromAccidentalDeletion $true
```

Querying Objects with Windows PowerShell

Parameter	Description
SearchBase	Defines the AD DS path to begin searching
SearchScope	Defines at what level below the SearchBase a search should be performed
ResultSetSize	Defines how many objects to return in response to a query
Properties	Defines which object properties to return and display
Filter	Defines a filter by using PowerShell syntax
LDAPFilter	Defines a filter by using LDAP query syntax

Descriptions of operators

-eq	Equal to	-gt	Greater than
-ne	Not equal to	-ge	Greater than or equal to
-lt	Less than	-like	Uses wildcards for pattern matching
-le	Less than or equal to		



Querying Objects with Windows PowerShell

Show all the properties for a user account:

```
Get-ADUser -Name "Administrator" -Properties *
```

Show all the user accounts in the Marketing OU and all its subcontainers:

```
Get-ADUser -Filter * -SearchBase "ou=Marketing,dc=adatum,dc=com" -SearchScope subtree
```

Show all of the user accounts with a last logon date older than a specific date:

```
Get-ADUser -Filter {lastlogondate -lt "January 1, 2012"}
```

Show all of the user accounts in the Marketing department that have a last logon date older than a specific date:

```
Get-ADUser -Filter {(lastlogondate -lt "January 1, 2012")} -and (department -eq "Marketing")}
```



Modifying Objects with Windows PowerShell

Use the pipe character (|) to pass a list of objects to a cmdlet for further processing

```
Get-ADUser -Filter {company -notlike "*"} | Set-ADUser -Company "A. Datum"
```

```
Get-ADUser -Filter {lastlogondate -lt "January 1, 2012"} | Disable-ADAccount
```

```
Get-Content C:\users.txt | Disable-ADAccount
```


Exercice

- Créez un compte utilisateur avec les propriétés suivantes : nom=Paul Dupont, SamAN=pdupont, UPN=pdupont@adatum.com, département=RH ;
- Créez 3 autres utilisateurs travaillant également dans le département RH ;
- Affichez tous les utilisateurs du département RH ;
- Créez un groupe global RH ;
- Créez une unité d'organisation Administration ;
- Ajoutez les utilisateurs et le groupe au sein de l'OU ;
- Affichez tous les objets de l'OU ;
- Ajoutez les utilisateurs au sein du groupe ;
- Affichez tous les utilisateurs du groupe ;
- Activez tous les comptes ;

Lesson 3: Performing Bulk Operations with Windows PowerShell

- What Are Bulk Operations?
- Demonstration: Using Graphical Tools to Perform Bulk Operations
- Demonstration: Performing Bulk Operations with Windows PowerShell

What Are Bulk Operations?

- Une opération « bulk » consiste à interagir avec plusieurs objets en une seule action.
- Exemples
 - Créer plusieurs utilisateurs en se basant d'un fichier texte.
 - Désactiver tous les comptes inactifs depuis 6 mois.
 - Renommer le département de plusieurs utilisateurs.
- Pour effectuer une opération « bulk » :
 - En utilisant l'interface graphique
 - En ligne de commandes
 - Script

Demonstration: Using Graphical Tools to Perform Bulk Operations

In this demonstration, you will see how to:

- Create a query for all users
- Configure the Company attribute for all users
- Verify that the Company attribute has been modified

Demonstration: Performing Bulk Operations with Windows PowerShell

In this demonstration, you will see how to:

- Configure a department for users
- Create an OU
- Run a script to create new user accounts
- Verify that new user accounts were created

Lab: Automating AD DS Administration by Using Windows PowerShell

- Exercise 1: Creating User Accounts and Groups by Using Windows PowerShell
- Exercise 2: Using Windows PowerShell to Create User Accounts in Bulk
- Exercise 3: Using Windows PowerShell to Modify User Accounts in Bulk

Logon Information

Virtual machines	20410D-LON-DC1 20410D-LON-CL1
User name	Adatum\Administrator
Password	Pa\$\$w0rd

Estimated Time: 45 minutes

Lab Scenario

You have been working for A. Datum Corporation for several years as a desktop support specialist. In this role, you visited desktop computers to troubleshoot app and network problems. You have recently accepted a promotion to the server support team. One of your first assignments is configuring the infrastructure service for a new branch office.

As part of configuring a new branch office, you need to create user and group accounts. Creating multiple users with graphical tools is inefficient, so, you will use Windows PowerShell.

Lab Review

- By default, are new user accounts enabled or disabled when you create them by using the **New-ADUser** cmdlet?
- What file extension do Windows PowerShell scripts use?

Module Review and Takeaways

- Review Questions
- Tools

